

OUTPUT

```
===== AI PROGRAM =====
```

1. Water Jug Problem
2. Mars Rover
3. 8 Queens Problem
4. OLA Cab Booking
5. Uniform Cost Search
6. Exit

```
Enter Choice: 1
```

```
--- Water Jug Problem ---
```

```
Step 1: Fill 3-gallon jug -> State (0, 3)
Step 2: Pour into 4-gallon jug -> State (3, 0)
Step 3: Fill 3-gallon jug again -> State (3, 3)
Step 4: Pour into 4-gallon jug until full -> State (4, 2)
Step 5: Empty 4-gallon jug -> State (0, 2)
Step 6: Pour remaining 2 gallons into 4-gallon jug -> State (2, 0)
```

```
Goal Achieved: 2 gallons in 4-gallon jug
```

```
===== AI PROGRAM =====
```

1. Water Jug Problem
2. Mars Rover
3. 8 Queens Problem
4. OLA Cab Booking
5. Uniform Cost Search
6. Exit

```
Enter Choice: 2
```

```
--- Mars Rover Intelligent Agent ---
```

```
Percepts:
```

- Camera Images
- Rock Samples
- Temperature
- Obstacles
- Battery Level

```
Actions:
```

- Move
- Collect Samples
- Capture Images
- Send Data
- Avoid Obstacles

```
Environment:
```

- Partially Observable
- Dynamic
- Sequential

```
Agent Type: Utility-Based Agent
```

```
===== AI PROGRAM =====
```

1. Water Jug Problem
2. Mars Rover
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5. Uniform Cost Search
6. Exit

Enter Choice: 3

--- 8 Queens Problem ---

```
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
```

===== AI PROGRAM =====

1. Water Jug Problem
2. Mars Rover
3. 8 Queens Problem
4. OLA Cab Booking
5. Uniform Cost Search
6. Exit

Enter Choice: 4

--- OLA Cab Booking ---

Enter Pickup Location: CHENNAI
Enter Destination: THANJAVUR

Available Cabs

1. Mini
2. Micro
3. Sedan
4. Shared
5. Prime

Choose Cab: 2

Booking Confirmed

Pickup: CHENNAI

Destination: THANJAVUR

Cab: Micro

===== AI PROGRAM =====

1. Water Jug Problem
2. Mars Rover
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5. Uniform Cost Search
6. Exit

Enter Choice: 5

--- Uniform Cost Search ---

Optimal Path: S -> A -> C -> G

Total Cost: 4

===== AI PROGRAM =====

1. Water Jug Problem
2. Mars Rover
3. 8 Queens Problem
4. OLA Cab Booking
5. Uniform Cost Search
6. Exit

Enter Choice: 6

Thank You

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